Math 110 — Week 4

Solutions

HW4 P1.

Idea. The identity in (a) hands you the inverse when $ad - bc \neq 0$, and forces $A = 0$ when $ad - bc = 0$ and A is assumed invertible.

1. (a) Write $B = \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$.

$$AB = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = (ad - bc)I.$$

Likewise

$$BA = \begin{pmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{pmatrix} = (ad - bc)I.$$

2. (b) If $\delta = ad - bc \neq 0$, then $A(\delta^{-1}B) = I = (\delta^{-1}B)A$. So A is invertible with

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

3. (c) Suppose $ad - bc = 0$ and A is invertible. Then $AB = 0$, and multiplying on the left by A^{-1} gives $B = 0$. So $d = b = c = a = 0$, i.e. $A = 0$. But the zero matrix is not invertible. Contradiction.

HW4 P2.

Idea. Compute $ad - bc$ in the stated field.

1. $\left(\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right)$: $6 - 5 = 1$. Invertible, inverse $\left(\begin{smallmatrix} 3 & -1 \\ -5 & 2 \end{smallmatrix}\right)$.
2. $\left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$: $(1)(-1) - i \cdot i = -1 + 1 = 0$. Not invertible. (Second column is i times the first.)
3. $\left(\begin{smallmatrix} 1 & 2 \\ 3 & 1 \end{smallmatrix}\right)$ over $\mathbb{F}_5$: $1 - 6 = -5 = 0$ in $\mathbb{F}_5$. Not invertible. (Second column is 2 times the first: $2 \cdot 3 = 6 = 1$.)

HW4 P3.

Idea. Injective maps keep independence; n independent vectors in dimension n are a basis. Conversely a basis in the range makes T surjective.

1. (a)⇒(b). Let $v_1, \ldots, v_n$ be a basis. If $\sum \lambda_j T v_j = 0$ then $T(\sum \lambda_j v_j) = 0$, so $\sum \lambda_j v_j = 0$ (T injective), so all $\lambda_j = 0$.
2. So $Tv_1, \ldots, Tv_n$ are n independent vectors in V, $\dim V = n$: a basis.
3. (b)⇒(a). Take any basis $v_1, \ldots, v_n$. Then $\operatorname{range} T \supseteq \operatorname{span}(Tv_1, \ldots, Tv_n) = V$, so T is surjective, hence an isomorphism (Prop. 31).

HW4 P4.

Idea. Rank is the dimension of the range. Compare the range of BC with the range of B, and view it as the image of $\operatorname{range} C$.

1. Every $BCx = B(Cx)$ is in $\operatorname{range} B$. So $\operatorname{range}(BC) \subseteq \operatorname{range} B$ and $\operatorname{rank}(BC) \leq \operatorname{rank} B$.
2. $\operatorname{range}(BC) = \{By : y \in \operatorname{range} C\}$ is the range of the restriction $B|_{\operatorname{range} C}$. By rank–nullity for this restriction, its dimension is $\dim \operatorname{range} C - \dim \operatorname{null}(B|_{\operatorname{range} C}) \leq \operatorname{rank} C$.
3. (b) Take $B = C = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: both have rank 1, and $BC = 0$ has rank 0.

HW4 P5.

Idea. Put each basis in its own slot.

1. Let $v_1, \dots, v_n$ and $w_1, \dots, w_m$ be bases of V and W. Claim: $(v_1, 0), \dots, (v_n, 0), (0, w_1), \dots, (0, w_m)$ is a basis of $V \times W$.
2. Spans:
$$(v, w) = (\sum a_i v_i, \sum b_j w_j) = \sum a_i (v_i, 0) + \sum b_j (0, w_j).$$
3. Independent:
$$\sum a_i (v_i, 0) + \sum b_j (0, w_j) = (\sum a_i v_i, \sum b_j w_j) = (0, 0)$$
gives $\sum a_i v_i = 0$ and $\sum b_j w_j = 0$, so all $a_i = 0$ and all $b_j = 0$.
4. So the list spans and is independent: it is a basis of $V \times W$. It has $n+m$ vectors, so $\dim(V \times W) = \dim V + \dim W$.

HW4 P6.

Idea. A subspace contains 0; that locates v.

1. (a) If $v + U$ is a subspace, $0 \in v + U$, so $0 = v + u$ and $v = -u \in U$.
2. If $v \in U$: $v + u \in U$ for all $u \in U$, so $v + U \subseteq U$; and each $u = v + (u - v) \in v + U$, so $U \subseteq v + U$. So $v + U = U$, a subspace.
3. (b) If $v + U = v' + U$, then $v = v + 0 \in v' + U$, so $v = v' + u$ and $v - v' \in U$.
4. If $v - v' \in U$: $v + u = v' + ((v - v') + u) \in v' + U$, so $v + U \subseteq v' + U$; since $v' - v \in U$ as well, the same argument with v, v' exchanged gives $v' + U \subseteq v + U$. So $v + U = v' + U$.

HW4 P7.

Idea. An invertible operator sends a basis to a basis. Build a basis of V starting in U, and a second basis starting with the images under S.

1. Let $u_1, \dots, u_m$ be a basis of U; extend to a basis $u_1, \dots, u_m, x_1, \dots, x_k$ of V.
2. $Su_1, \dots, Su_m$ is independent since S is injective. Extend it to a basis $Su_1, \dots, Su_m, y_1, \dots, y_k$ of V (the same k, since both bases have $\dim V$ vectors).
3. Define T by $Tu_i = Su_i$, $Tx_j = y_j$. Then T and S agree on a basis of U, so $T|_U = S$.
4. $\operatorname{range} T$ is a subspace containing the basis $Su_1, \dots, Su_m, y_1, \dots, y_k$ of V, hence all of V: T is surjective, hence invertible (Prop. 31).

R1.

Idea. S must undo T on $\operatorname{range} T$; what it does elsewhere is free. Use a basis of W that starts with $Tv_1, \dots, Tv_n$.

1. (a) If $\sum \lambda_j T v_j = 0$ then $T(\sum \lambda_j v_j) = 0$, so $\sum \lambda_j v_j = 0$, so all $\lambda_j = 0$.
2. (b) Extend $Tv_1, \dots, Tv_n$ to a basis $Tv_1, \dots, Tv_n, w_1, \dots, w_k$ of W. Define S by $S(Tv_j) = v_j$ and $S(w_i) = 0$.
3. Then $ST(v_j) = v_j$ for each j: ST and I_V agree on a basis, so $ST = I_V$.

R2.

Idea. This is the map Γ of Prop. 37 for two subspaces; directness is exactly injectivity.

1. $\Gamma(u, w) = u + w$ is linear, because the operations in $U \times W$ are componentwise.
2. Γ maps onto $U + W$: every element of $U + W$ is $u + w$ with $u \in U$, $w \in W$.
3. Γ is injective: if $u + w = 0$, then $u = -w \in U \cap W = \{0\}$, so $u = w = 0$.
4. By Prop. 29, $\Gamma \colon U \times W \to U \oplus W$ is an isomorphism.

R3.

Idea. Injectivity is checked on the null space; for surjectivity, find a sequence that is not an output: every output starts with 0.

1. (i) If $R(a) = (0, a_1, a_2, \dots) = 0$, then every $a_j = 0$. So $\operatorname{null} R = \{0\}$ and R is injective.
2. Every output of R has first entry 0, so $(1, 0, 0, \dots)$ is not in the range: R is not surjective.
3. (ii) For an operator on a finite-dimensional space, injective implies surjective (Prop. 31). So no such operator is injective and not surjective.

R4.

Idea. Split v as $(v - Pv) + Pv$.

1. $Pv \in \operatorname{range} P$, and $P(v - Pv) = Pv - P^2v = 0$, so $v - Pv \in \operatorname{null} P$. So every $v = (v - Pv) + Pv$ is in $\operatorname{null} P + \operatorname{range} P$; and $\operatorname{null} P + \operatorname{range} P \subseteq V$. Hence $V = \operatorname{null} P + \operatorname{range} P$.
2. If $v \in \operatorname{null} P \cap \operatorname{range} P$, write $v = Pw$. Then $v = Pw = P^2w = Pv = 0$, because $v \in \operatorname{null} P$. So $\operatorname{null} P \cap \operatorname{range} P = \{0\}$, and with step 1, $V = \operatorname{null} P \oplus \operatorname{range} P$ (Def. 6).

R5.

Idea. The matrix says $D\beta_1 = \gamma_1$, $D\beta_2 = \gamma_2$, $D\beta_3 = \gamma_3$, $D\beta_4 = 0$. Put the null space last and let γ be the images.

1. Column 4 of the matrix is 0, so $D\beta_4 = 0$. $\operatorname{null} D$ is the constants, so take $\beta_4 = 1$.
2. Take $\beta_1, \beta_2, \beta_3 = x, x^2, x^3$ and set $\gamma_j = D\beta_j$: $\gamma = (1,\ 2x,\ 3x^2)$.
3. Both $\beta = (x, x^2, x^3, 1)$ and γ consist of polynomials of distinct degrees, so each is independent. β has $4 = \dim \mathcal{P}_3(\mathbb{R})$ vectors and γ has $3 = \dim \mathcal{P}_2(\mathbb{R})$, so each is a basis (W2).
4. Columns: $D\beta_j = \gamma_j$ is column j equal to e_j for $j \le 3$, and $D\beta_4 = 0$ is column 4 equal to 0. This is the given matrix.

R6.

Idea. An isomorphism has zero null space and full range; rank–nullity converts that into dimensions.

1. Let $T\colon V \to W$ be an isomorphism. It is injective, so $\operatorname{null} T = \{0\}$, and surjective, so $\operatorname{range} T = W$ (Prop. 29).
2. Rank–nullity: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T = 0 + \dim W$.

R7.

Idea. A vector of $\operatorname{null} T$ is a combination of the basis; T sends it to the same combination of the Tv_j.

1. **Assume T is an isomorphism.** If $\sum \lambda_j Tv_j = 0$, then $T(\sum \lambda_j v_j) = 0$, so $\sum \lambda_j v_j = 0$ because T is injective, and all $\lambda_j = 0$ because the v_j are independent.
2. **Assume $Tv_1, \dots, Tv_n$ is independent.** Let $v = \sum \lambda_j v_j \in \operatorname{null} T$. Then $0 = Tv = \sum \lambda_j Tv_j$, so all $\lambda_j = 0$ and $v = 0$. So T is injective.
3. T is an operator on a finite-dimensional space, so it is an isomorphism (Prop. 31).